

wrangle_report

July 23, 2024

0.1 Reporting: wrangle_report

- Create a **300-600 word written report** called "wrangle_report.pdf" or "wrangle_report.html" that briefly describes your wrangling efforts. This is to be framed as an internal document.

0.2 Introduction

- In this project, I wrangled and analyzed the WeRateDogs Twitter data to create interesting and trustworthy analyses and visualizations.
- The goal is to collect data from multiple sources, extract quality and ranking problems from them, solve them, and extract insights from visualisation.

0.3 Data Gathering

0.3.1 WeRateDogs Twitter Archive

- 'twitter-archive-enhanced.csv' was provided and downloaded manually as a CSV file.

0.3.2 Image predictions file

- I downloaded the image predictions file programmatically using the Requests library from the provided URL.

0.3.3 Tweet json file

- I downloaded the Tweet json file and read it by function read_json().

0.4 Data Assessing

0.4.1 Visual Assessment

- I read the data, took a look at it, and began to see problems visually.

0.4.2 Programmatic Assessment

- I used pandas functions like value_counts(), info(), describe(), sort_values() and duplicated() To discover problems in the data.

0.4.3 Identified Issues

- **Quality Issues:**

1. Erroneous datatypes in 'timestamp' in df_twitter_archive table
2. There are illogical evaluations in 'rating_numerator' in df_twitter_archive table
3. There are illogical evaluations in 'rating_denominator' in df_twitter_archive table
4. There are illogical names in 'name' in df_twitter_archive table because of 'text'
5. "contributors", "coordinates", "geo-location", "quoted_status", etc. have a lot of missing values or are empty in the df_tweet_json table.
6. Some predictions are not dogs (e.g. box_turtle, shopping_cart, hen, desktop_computer, three-toed_sloth, convertible) in df_image_predictions table
7. There are duplicates in 'jpg_url' in df_image_predictions table
8. There are urls wrong in columns 'expanded_urls' in df_twitter_archived table
9. There is illogical value in 'p1_conf' in row 106 in df_image_predictions table

- **Tidiness Issues:**

1. df_image_predictions and df_tweet_json should be part of the df_twitter_archive table
2. Columns 'retweeted_status' and 'quoted_status' are redundant because we are only interested in original tweets.
3. Column name difference 'id' in df_tweet_json table from 'tweet_id'

0.5 Data Cleaning

0.5.1 Define

- For each identified issue, I explain how the problem will be solved.

0.5.2 Code

- I implement what you explained in Define using code.

0.5.3 Test

- I am checking whether what I did in the code actually solved the problem or not

```
"""python ### Example cleaning code ### Define : Convert timestamp to
datetime data type.      ### Code : * df_twitter_archive_clean['timestamp'] =
pd.to_datetime(df_twitter_archive_clean['timestamp'] , errors='coerce') ### Test : *
df_twitter_archive_clean.info()
```